SUPPLEMENTARY MATERIAL

Habitual physical activity and motor cortical neurophysiology: A systematic review with meta-analyses of transcranial magnetic stimulation measures

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Supplementary Material 1. Search strategies for all databases

Supplementary Material 2. R code for calculating Pearson's R and exact p-values

Supplementary Material 3. Subgroup analyses comparing self-reported and device-based physical activity measures

Supplementary Material 1. Search strategies for all databases

PubMed

(((((((("Physical activity"[Title/Abstract]) OR ("Exercise"[Title/Abstract])) OR ("physical education"[Title/Abstract])) OR ("training"[Title/Abstract])) OR ("cycling"[Title/Abstract])) OR ("treadmill"[Title/Abstract])) OR ("single bout"[Title/Abstract]) AND ((clinicalstudy[Filter] OR clinicaltrial[Filter] OR observationalstudy[Filter] OR randomizedcontrolledtrial[Filter]) AND (2000:2025[pdat]))) AND ((((((("TMS"[Title/Abstract]) OR ("transcranial magnetic stimulation"[Title/Abstract])) OR ("motor evoked potential"[Title/Abstract])) OR ("stimulus-response curve"[Title/Abstract])) OR ("input-output curve"[Title/Abstract])) OR ("intracortical inhibition"[Title/Abstract])) OR ("intracortical facilitation"[Title/Abstract])) OR ("silent period"[Title/Abstract]) AND ((clinicalstudy[Filter] OR clinicaltrial[Filter] OR observationalstudy[Filter] OR randomizedcontrolledtrial[Filter] OR randomizedcontrolledtrialveterinary[Filter] OR veterinaryobservationalstudy[Filter])))

PsycINFO

1. Physical Activity
2. "exercise".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
3. "physical education".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
4. "training".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
5. "cycling".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
6. "treadmill".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
7. "single bout".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
8. Transcranial Magnetic Stimulation/
9. "TMS".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
10. "motor-evoked potential".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
11. "stimulus-response curve".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
12. "input-output curve".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
13. "intracortical inhibition".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
14. "intracortical facilitation".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
15. "silent period".mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]
16. 1 or 2 or 3 or 4 or 5 or 6 or 7
17. 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15
18. 16 and 17

SPORTDiscus

S1: AB "physical activity" OR AB "exercise" OR AB "physical education" OR AB "training" OR AB "Cycling" OR AB "single bout" OR AB "treadmill"

S2: AB "transcranial magnetic stimulation" OR AB "TMS" OR AB "motor evoked potential" OR AB "input-output curve" OR AB "Stimulus-response curve" OR AB "intracortical inhibition" OR AB "intracortical facilitation" OR XB "silent period"

S3: (S1 AND S2)

Web of Science

1: ((((((AB=("physical activity")) OR AB=("exercise")) OR AB=("physical education")) OR AB=("training")) OR AB=("cycling")) OR AB=("treadmill")) OR AB=("single bout")

2: (((((((AB=("TMS")) OR AB=("Transcranial Magnetic stimulation")) OR AB=("motor evoked potential")) OR AB=("stimulus-response curve")) OR AB=("input-output curve")) OR AB=("intracortical inhibition")) OR AB=("intracortical facilitation")) OR AB=("silent period")

3: #2 AND #1

CINAHL

S1: AB "physical activity" OR AB "exercise" OR AB "training" OR AB "physical education" OR AB "Cycling" OR AB "treadmill" OR AB "single bout"

S2: AB "transcranial magnetic stimulation" OR AB "TMS" OR AB "motor evoked potential" OR AB "input-output curve" OR AB "Stimulus-response curve" OR AB "intracortical inhibition" OR AB "intracortical facilitation" OR XB "silent period"

S3: (S1 AND S2)

Supplementary Material 2. R code for calculating Pearson's R and exact p-values

2.1. Calculation of Pearson's r from the exact p-value and degrees of freedom

R code for calculating the Pearson correlation coefficient corresponding to a specified exact p-value and degrees of freedom, when the direction of the relationship is known (Ripley et al., 2014; Cohen et al., 2002).

```
# Required libraries
library (MASS)
#Target p-value
target_p_value<- 0.0009
#Degrees of freedom (sample size-2)
df<- 30
#Calculate the critical value for t
t_critical<- qt(1-target_p_value/2, df=df)
#Convert t to r
r_critical<- sqrt(t_critical^2/(t_critical^2+df))
#Print the critical value for r
print(r_critical)
```

2.2. Calculation of the exact p-value from Pearson's r and sample size

R code for calculating the exact two-sided p-value associated with a Pearson correlation coefficient based on the sample size (Cohen et al., 2002).

```
# Sample size n<- 45
#Pearson correlation coefficient r<-0.25
#Compute the t-statistic
t_statistic<- r*sqrt((n-2)/(1-r^2))
#Degrees of freedom
df<- n-2
p_value <- 2 * pt(-abs(t_statistic), df = df)
```

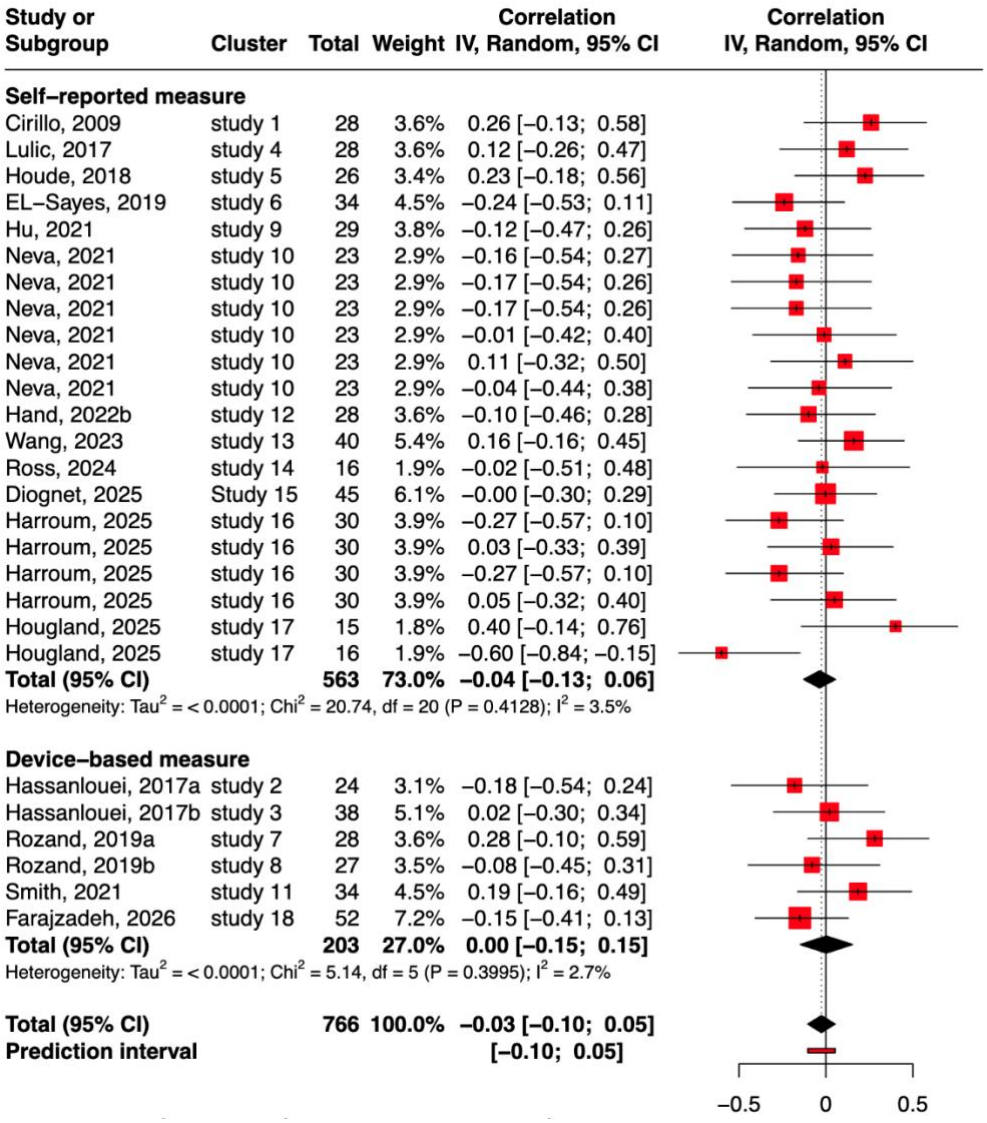
References

-Ripley B, Venables B, Bates DM, et al. MASS: support functions and datasets for Venables and Ripley's MASS. Version 7.3-61 [R package]. 2024. Available from: https://cran.r-project.org/web/packages/MASS/MASS.pdf.

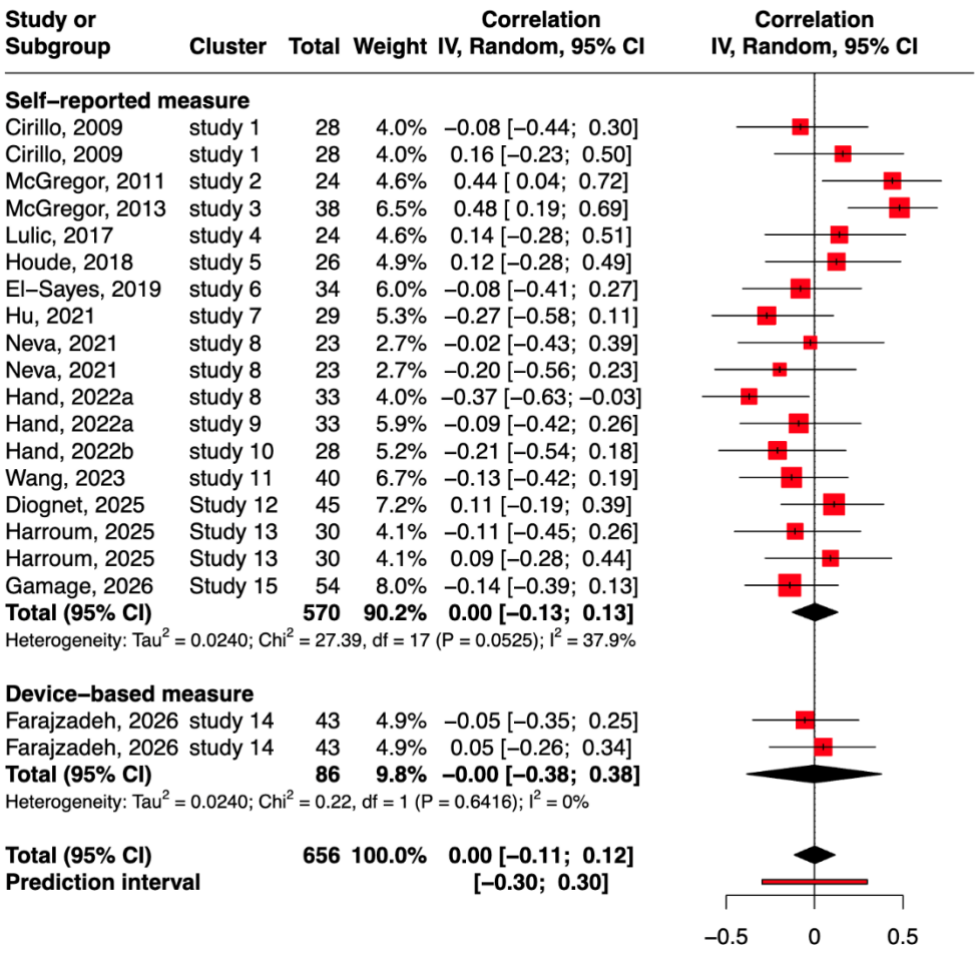
-Cohen J, Cohen P, West SG, et al. Applied multiple regression/correlation analysis for the behavioral sciences. 3rd ed. Mahwah, NJ: Lawrence Erlbaum Associates; 2002.

Supplementary Material 3. Subgroup analyses comparing self-reported and device-based physical activity measures

3.1. Corticospinal Excitability



3.2. Inhibitory Processes



3.3. Facilitatory Processes

